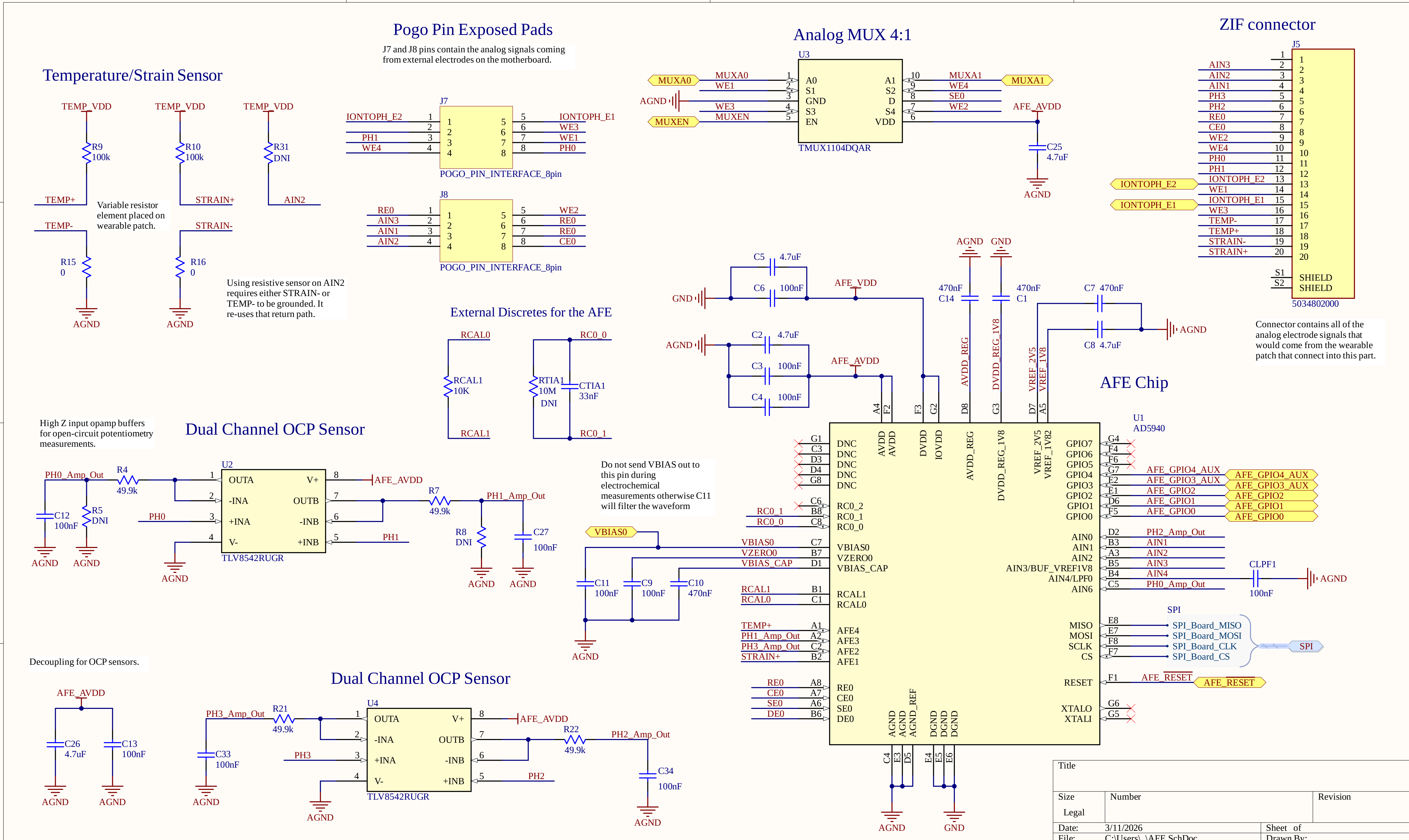

Temperature/Strain Sensor
TEMP_VDD
R9 100k
R10 100k
R31 DNI
TEMP+
STRAIN+
AIN2
TEMP-
STRAIN-
R15 0
R16 0
AGND
Variable resistor element placed on wearable patch.
Using resistive sensor on AIN2 requires either STRAIN- or TEMP- to be grounded. It re-uses that return path.
Pogo Pin Exposed Pads
J7 and J8 pins contain the analog signals coming from external electrodes on the motherboard.
J7
IONTOPH_E2
PH1
WE4
IONTOPH_E1
WE3
WE1
PH0
POGO_PIN_INTERFACE_8pin
J8
RE0
AIN3
AIN1
AIN2
WE2
CE0
POGO_PIN_INTERFACE_8pin
External Discretes for the AFE
RCAL0
RCAL1 10K
RCAL1
RC0_0
RTIA1 10M DNI
CTIA1 33nF
RC0_1
Analog MUX 4:1
U3
MUXA0
MUXA1
MUXEN
AGND
A0 S1 GND S3 EN
A1 S2 D S4 VDD
WE1
WE4
SE0
WE3
WE2
AFE_AVDD
TMUX1104DQAR
C25 4.7uF
ZIF connector
J5
AIN3
AIN2
AIN1
PH3
PH2
RE0
CE0
WE2
WE4
PH0
PH1
IONTOPH_E2
WE1
IONTOPH_E1
WE3
TEMP-
TEMP+
STRAIN-
STRAIN+
SHIELD
SHIELD
5034802000
Connector contains all of the analog electrode signals that would come from the wearable patch that connect into this part.
C5 4.7uF
C6 100nF
AFE_VDD
GND
C2 4.7uF
C3 100nF
C4 100nF
AFE_AVDD
AGND
GND
470nF C14
470nF C1
C7 470nF
C8 4.7uF
AVDD_REG
DVDD_REG_1V8
VREF_2V5
VREF_1V8
AFE Chip
U1
AD5940
DNC
AVDD
DVDD
IOVDD
AVDD_REG
DVDD_REG_1V8
VREF_2V5
VREF_1V82
GPIO7 GPIO6 GPIO5 GPIO4 GPIO3 GPIO2 GPIO1 GPIO0
AFE_GPIO4_AUX
AFE_GPIO3_AUX
AFE_GPIO2
AFE_GPIO1
AFE_GPIO0
RC0_2 RC0_1 RC0_0
VBIAS0 VZERO0 VBIAS_CAP
VBIAS0
VZERO0
VBIAS_CAP
RCAL1 RCAL0
AFE4 AFE3 AFE2 AFE1
TEMP+
PH1_Amp_Out
PH3_Amp_Out
STRAIN+
RE0 CE0 SE0 DE0
AGND AGND AGND_REF
DGND DGND DGND
AIN0 AIN1 AIN2 AIN3/BUF_VREF1V8 AIN4/LPF0 AIN6
PH2_Amp_Out
AIN1
AIN2
AIN3
AIN4
PH0_Amp_Out
CLPF1 100nF
MISO MOSI SCLK CS
SPI
SPI_Board_MISO
SPI_Board_MOSI
SPI_Board_CLK
SPI_Board_CS
RESET
AFE_RESET
XTALO XTALI
Do not send VBIAS out to this pin during electrochemical measurements otherwise C11 will filter the waveform
C11 100nF
C9 100nF
C10 470nF
High Z input opamp buffers for open-circuit potentiometry measurements.
Dual Channel OCP Sensor
U2
PH0_Amp_Out
R4 49.9k
C12 100nF
R5 DNI
PH0
OUTA -INA +INA V-
V+ OUTB -INB +INB
AFE_AVDD
TLV8542RUGR
R7 49.9k
PH1_Amp_Out
R8 DNI
C27 100nF
PH1
Decoupling for OCP sensors.
AFE_AVDD
C26 4.7uF
C13 100nF
Dual Channel OCP Sensor
U4
PH3_Amp_Out
R21 49.9k
C33 100nF
PH3
TLV8542RUGR
R22 49.9k
PH2_Amp_Out
C34 100nF
PH2
Title
Size Legal
Number
Revision
Date: 3/11/2026
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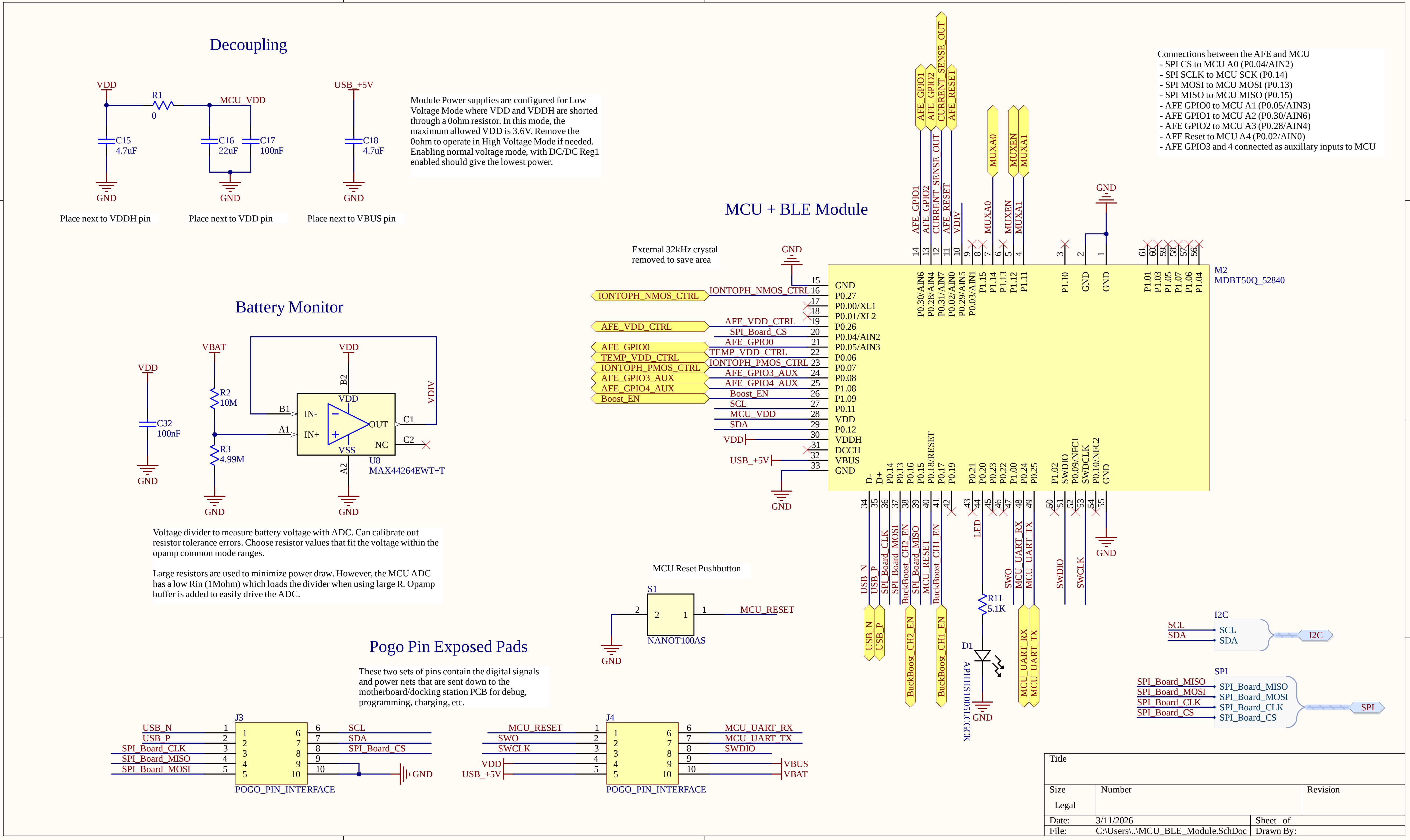
Decoupling
VDD
R1
0
MCU_VDD
C15
4.7uF
C16
22uF
C17
100nF
USB_+5V
C18
4.7uF
GND
Place next to VDDH pin
Place next to VDD pin
Place next to VBUS pin
Module Power supplies are configured for Low Voltage Mode where VDD and VDDH are shorted through a 0ohm resistor. In this mode, the maximum allowed VDD is 3.6V. Remove the 0ohm to operate in High Voltage Mode if needed. Enabling normal voltage mode, with DC/DC Reg1 enabled should give the lowest power.
Connections between the AFE and MCU
- SPI CS to MCU A0 (P0.04/AIN2)
- SPI SCLK to MCU SCK (P0.14)
- SPI MOSI to MCU MOSI (P0.13)
- SPI MISO to MCU MISO (P0.15)
- AFE GPIO0 to MCU A1 (P0.05/AIN3)
- AFE GPIO1 to MCU A2 (P0.30/AIN6)
- AFE GPIO2 to MCU A3 (P0.28/AIN4)
- AFE Reset to MCU A4 (P0.02/AIN0)
- AFE GPIO3 and 4 connected as auxillary inputs to MCU
MCU + BLE Module
External 32kHz crystal removed to save area
M2
MDBT50Q_52840
AFE_GPIO1
AFE_GPIO2
CURRENT_SENSE_OUT
AFE_RESET
VDIV
MUXA0
MUXEN
MUXA1
IONTOPH_NMOS_CTRL
AFE_VDD_CTRL
SPI_Board_CS
AFE_GPIO0
TEMP_VDD_CTRL
IONTOPH_PMOS_CTRL
AFE_GPIO3_AUX
AFE_GPIO4_AUX
Boost_EN
SCL
MCU_VDD
SDA
VDD
USB_+5V
GND
P0.27
P0.00/XL1
P0.01/XL2
P0.26
P0.04/AIN2
P0.05/AIN3
P0.06
P0.07
P0.08
P1.08
P1.09
P0.11
VDD
P0.12
VDDH
DCCH
VBUS
P0.30/AIN6
P0.28/AIN4
P0.31/AIN7
P0.02/AIN0
P0.29/AIN5
P0.03/AIN1
P1.15
P1.14
P1.13
P1.12
P1.11
P1.10
P1.01
P1.03
P1.05
P1.07
P1.06
P1.04
D-
D+
P0.14
P0.13
P0.16
P0.15
P0.18/RESET
P0.17
P0.19
P0.21
P0.20
P0.23
P0.22
P1.00
P0.24
P0.25
P1.02
SWDIO
P0.09/NFC1
SWDCLK
P0.10/NFC2
USB_N
USB_P
SPI_Board_CLK
SPI_Board_MOSI
BuckBoost_CH2_EN
SPI_Board_MISO
MCU_RESET
BuckBoost_CH1_EN
LED
SWO
MCU_UART_RX
MCU_UART_TX
SWCLK
R11
5.1K
D1
APHHS1005LCGCK
Battery Monitor
VBAT
R2
10M
R3
4.99M
C32
100nF
IN-
IN+
OUT
NC
VSS
U8
MAX44264EWT+T
Voltage divider to measure battery voltage with ADC. Can calibrate out resistor tolerance errors. Choose resistor values that fit the voltage within the opamp common mode ranges.
Large resistors are used to minimize power draw. However, the MCU ADC has a low Rin (1Mohm) which loads the divider when using large R. Opamp buffer is added to easily drive the ADC.
MCU Reset Pushbutton
S1
NANOT100AS
I2C
SPI
Pogo Pin Exposed Pads
These two sets of pins contain the digital signals and power nets that are sent down to the motherboard/docking station PCB for debug, programming, charging, etc.
J3
POGO_PIN_INTERFACE
J4
POGO_PIN_INTERFACE
SWO
SWCLK
SWDIO
VBUS
VBAT
Title
Size
Legal
Number
Revision
Date: 3/11/2026
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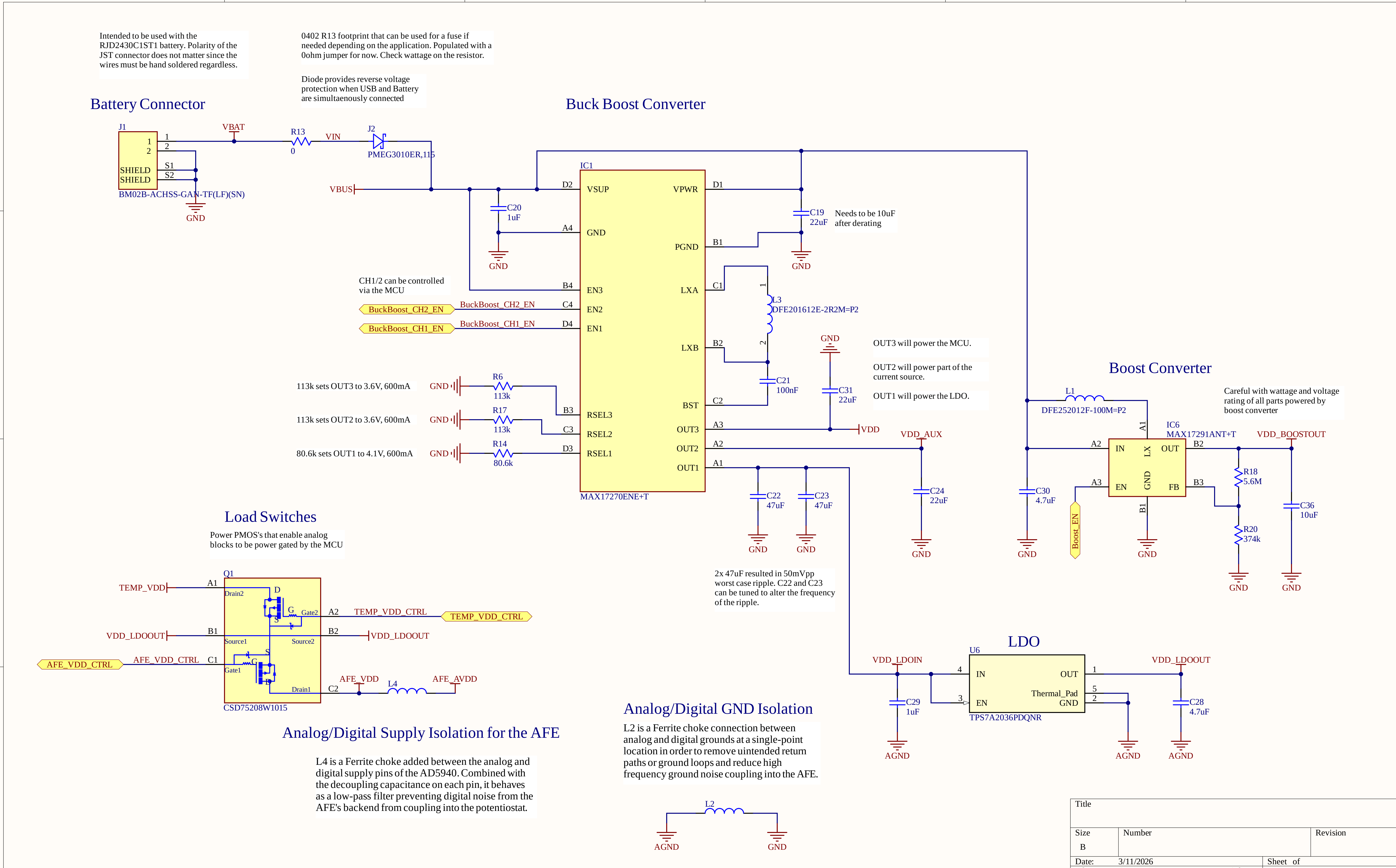

Intended to be used with the RJD2430C1ST1 battery. Polarity of the JST connector does not matter since the wires must be hand soldered regardless.
0402 R13 footprint that can be used for a fuse if needed depending on the application. Populated with a 0ohm jumper for now. Check wattage on the resistor.
Diode provides reverse voltage protection when USB and Battery are simultaenously connected
Battery Connector
J1
BM02B-ACHSS-GAN-TF(LF)(SN)
SHIELD
SHIELD
S1
S2
GND
VBAT
R13
0
VIN
J2
PMEG3010ER,115
VBUS
Buck Boost Converter
IC1
MAX17270ENE+T
VSUP
GND
EN3
EN2
EN1
RSEL3
RSEL2
RSEL1
VPWR
PGND
LXA
LXB
BST
OUT3
OUT2
OUT1
C20
1uF
C19
22uF
Needs to be 10uF after derating
L3
DFE201612E-2R2M=P2
C21
100nF
C31
22uF
CH1/2 can be controlled via the MCU
BuckBoost_CH2_EN
BuckBoost_CH1_EN
113k sets OUT3 to 3.6V, 600mA
113k sets OUT2 to 3.6V, 600mA
80.6k sets OUT1 to 4.1V, 600mA
R6
113k
R17
113k
R14
80.6k
VDD
VDD_AUX
OUT3 will power the MCU.
OUT2 will power part of the current source.
OUT1 will power the LDO.
C22
47uF
C23
47uF
C24
22uF
2x 47uF resulted in 50mVpp worst case ripple. C22 and C23 can be tuned to alter the frequency of the ripple.
Boost Converter
Careful with wattage and voltage rating of all parts powered by boost converter
L1
DFE252012F-100M=P2
IC6
MAX17291ANT+T
IN
EN
LX
GND
OUT
FB
VDD_BOOSTOUT
Boost_EN
C30
4.7uF
R18
5.6M
R20
374k
C36
10uF
Load Switches
Power PMOS's that enable analog blocks to be power gated by the MCU
Q1
CSD75208W1015
Drain2
Gate2
Source1
Source2
Gate1
Drain1
TEMP_VDD
TEMP_VDD_CTRL
VDD_LDOOUT
AFE_VDD_CTRL
AFE_VDD
L4
AFE_AVDD
LDO
U6
TPS7A2036PDQNR
IN
EN
OUT
Thermal_Pad
GND
VDD_LDOIN
VDD_LDOOUT
C29
1uF
C28
4.7uF
AGND
Analog/Digital Supply Isolation for the AFE
L4 is a Ferrite choke added between the analog and digital supply pins of the AD5940. Combined with the decoupling capacitance on each pin, it behaves as a low-pass filter preventing digital noise from the AFE's backend from coupling into the potentiostat.
Analog/Digital GND Isolation
L2 is a Ferrite choke connection between analog and digital grounds at a single-point location in order to remove unintended return paths or ground loops and reduce high frequency ground noise coupling into the AFE.
L2
AGND
GND
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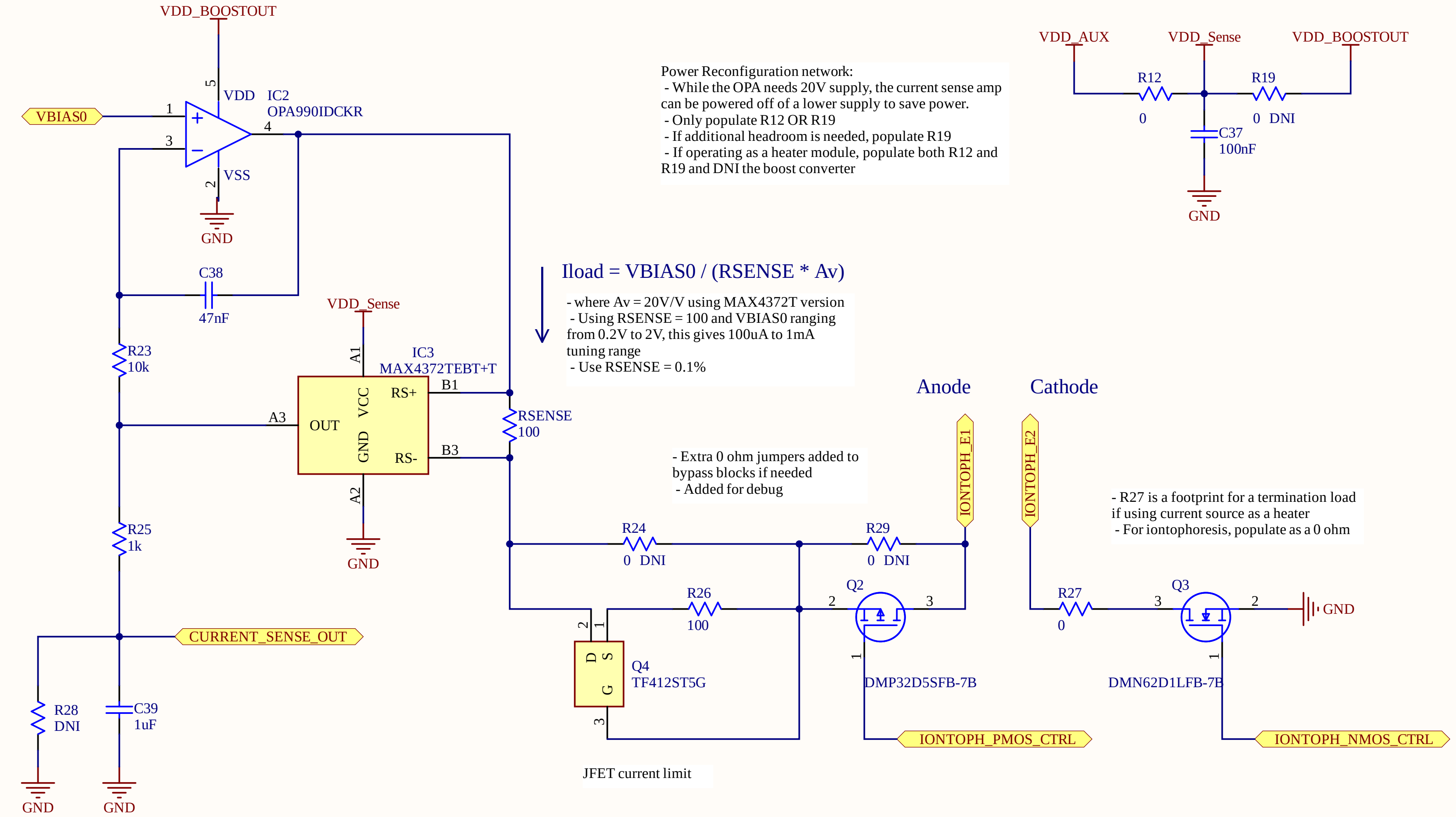
Voltage Controlled Current Source



Power Reconfiguration network:
- While the OPA needs 20V supply, the current sense amp can be powered off of a lower supply to save power.
- Only populate R12 OR R19
- If additional headroom is needed, populate R19
- If operating as a heater module, populate both R12 and R19 and DNI the boost converter

Iload = VBIAS0 / (RSENSE * Av)

- where Av = 20V/V using MAX4372T version
- Using RSENSE = 100 and VBIAS0 ranging from 0.2V to 2V, this gives 100uA to 1mA tuning range
- Use RSENSE = 0.1%

- Extra 0 ohm jumpers added to bypass blocks if needed
- Added for debug

- R27 is a footprint for a termination load if using current source as a heater
- For iontophoresis, populate as a 0 ohm

JFET current limit

- LPF before going into MCU ADC for monitoring the current
- Note: Current sense amp has very low output impedance
- Footprint added for resistive divider in case the sense amp needs larger VDD (prevents MCU from seeing anything larger than 3.9V)

Title			
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